Predicting Electricity Spikes: Evidence from Supply, Demand and Renewable Conditions

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Purpose

This project aims to study Alberta's electricity market, where scarcity, demand surges, renewable variability, and system conditions can produce extreme hourly pool prices. The research question is: Can electricity price spike hours be predicted using publicly available system-level data on demand, supply conditions, and renewable generation?

Accurate spike prediction has direct operational value for grid operators, energy retailers, and large industrial consumers scheduling energy-intensive processes. Since Alberta is a deregulated market, this is an interesting and relevant problem, as spikes are frequent and driven by interactions between demand, renewables availability, and market capacity.

Data

Our analysis uses two publicly available datasets from the Alberta Electric System Operator (AESO).

1. Hourly Metered Volumes and Pool Price and AIL file (Alberta Electric System Operator, 2025a), which provides hourly pool price (CAD/MWh), Alberta Internal Load, and electricity import/export flows.
2. Historical Generation Data (CSD) (Alberta Electric System Operator, 2025b), which records individual generating unit outputs with fuel type labels, containing province-wide wind and solar totals.

Both are merged at the hour level, producing 47K observations covering 2020–2025. The main preprocessing challenge is that the CSD files require aggregation before merging. Minor datetime inconsistencies around daylight saving transitions will be resolved using the Mountain Prevailing Time column present in both files. Lagged and moving average features need to be included. For exploratory analysis, we plan a price distribution plot to visualize the spike tail, a heatmap of average prices by hour and month, time series plots of energy generation by fuel type alongside pool price, as well as a correlation matrix of input features.

Topic to Investigate

The Alberta Power Pool operates on a dispatch system where the hourly price is determined by the intersection of supply and demand. Price spikes typically occur when the supply-demand balance tightens significantly, often due to a combination of high demand, unplanned outages, or low renewable output. This project extends the literature for Alberta Energy Market by applying neural network models to spike classification using data from 2020–2025.

Previous studies have documented Alberta's electricity price volatility driven by its deregulated structure, reliance on natural gas, and growing renewable capacity. Our project follows the approach taken by *Manfre Jaimes et al.* (2023), from the University of Calgary, who demonstrated that ML models can effectively forecast multi-day-ahead prices in this specific market, using neural networks (LSTM) and XGBoost. *Zamudio López et al.* (2024) directly tackle price spike classification in Alberta using Decision trees, reporting F1-scores around 66% with interpretable rule-based models. *Saha C.* (2025) developed a Grid Risk Assessment Tool from all North American market data, using logistic regression, XGBoost and random forests to predict instability events over 2019–2024. *Lago et al.* (2018) provided a benchmark comparison of 27 forecasting approaches and showed that deep learning methods consistently outperform statistical baselines on electricity price data, motivating our neural network approach.

Our study extends *Manfre Jaimes et al.* (2023) work by applying and comparing an MLP and a CNN to predict price spikes on a multi-hour-ahead basis, using the 2020–2025 period. A price spike will be flagged when an hourly pool price exceeded $200/MWh.

Statistical Methods

We will apply two binary classification models to determine which better predicts Alberta electricity price spikes. The first is a Multilayer Perceptron (MLP), which will take a flat feature vector of hourly market inputs. The second is a Convolutional Neural Network (CNN), which will leverage the temporal structure of the data to detect patterns across multiple days. Both models are feasible to implement in Python and will be trained using the *TimeSeriesSplit* procedure from Scikit-learn, which preserves temporal order by splitting data into sequential training and test folds. Performance will be compared using F1-score as the primary metric given class imbalance, supplemented by precision, recall, and ROC-AUC.

References

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